Section 1 Disorders of the Alimentary Tract

332 Approach to the Patient with Gastrointestinal Disease

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ANATOMIC CONSIDERATIONS

The gastrointestinal (GI) tract extends from the mouth to the anus and is composed of organs with distinct functions. Sphincters assist in gut compartmentalization to separate the organs. Distinct gut wall layers contribute to regional activities. The mucosa regulates fluid and nutrient transfer but is also a barrier to prevent entry of some luminal contents. Smooth muscle and enteric nerves mediate propulsion. Many organs possess a serosal layer to provide a supportive foundation and permit external input. The gut interacts with many other systems. Pancreaticobiliary conduits deliver bile and enzymes into the duodenum. Lymphatic channels assist in gut immune activities. Intrinsic nerves control propulsion and fluid regulation. Extrinsic neural input provides volitional or involuntary control specific for each gut region.

FUNCTIONS OF THE GI TRACT

The GI tract serves two main functions—assimilating nutrients and eliminating waste. In the mouth, food is processed, mixed with salivary amylase, and delivered to the pharynx. Aborally propagating esophageal contractions propel the ingested bolus into the stomach in coordination with relaxation of the upper and lower esophageal sphincters. Basal tone in the lower esophageal sphincter prevents gastroesophageal reflux.

The stomach triturates and mixes food with pepsin and acid. The proximal stomach serves a storage function by relaxing to accommodate the meal. Phasic contractions in the distal stomach propel food residue against the pylorus, where it is ground and thrust proximally for further mixing prior to emptying into the duodenum.

Most nutrient absorption occurs in the small intestine. Mucosal villi provide maximal surface area and possess specialized enzymes and transporters. Triturated food from the stomach mixes with pancreatic secretions and bile in the duodenum. Pancreatic juice contains both enzymes for digestion and bicarbonate to optimize luminal pH for enzyme activation. Bile secreted by the liver and stored in the gallbladder is needed for lipid digestion. Most nutrients and minerals are rapidly absorbed in the proximal intestine, while bile acids and vitamin B_{12} are absorbed in the ileum. Bile contains byproducts of erythrocyte degradation, toxins, medications, and cholesterol that are emptied into the colon along with indigestible residue. The ileocecal junction is a sphincter that limits coloileal reflux of colonic microbes.

The colon prepares waste for evacuation by to-and-fro contractions that dehydrate the stool. As a consequence, daily luminal volumes decrease from 1000–1500 mL in the ileum to 100–200 mL that are expelled from the rectum. Esophageal transit takes seconds, stomach and small intestine emptying times range from minutes to a few hours, but colon propagation requires >1 day in most individuals. A dense colonic microbiome ferments undigested carbohydrates and short-chain fatty acids and modulates immune activity. The colon terminates in the anus, which possesses volitional and involuntary controls to retain stool in the rectum until it can be released in a convenient setting.

EXTRINSIC MODULATION OF GUT FUNCTION

GI function is modified by influences outside the gut. Unlike other systems, the gut is in continuity with the outside environment. Several mechanisms protect against injury from foods, medications, toxins, and microbes. Mucosal immune pathways include epithelial and lamina propria lymphocytes and plasma cells and lymph node chains to prevent noxious agents from entering the circulation. Antimicrobial peptides secreted by Paneth cells defend against pathogens. Drugs and toxins absorbed into the portal venous circulation are filtered and detoxified in the liver. Many GI reflexes involve extrinsic vagus or splanchnic nerve pathways. The brain-gut axis alters function in gut regions not under volitional regulation. Stress disrupts normal gut motor, secretory, and sensory activities.

OVERVIEW OF GI DISEASES

GI diseases range in severity from conditions that produce mild symptoms and no long-term morbidity to those with intractable symptoms or adverse outcomes. Diseases may be localized to one gut organ or exhibit diffuse involvement at many sites.

CLASSIFICATION OF GI DISEASES

GI diseases are manifestations of alterations in nutrient and fluid assimilation, transit and sensation, immune function, and vascular supply. Neoplastic degeneration may affect any GI site. Genetic factors underlie some disorders.

Impaired Digestion and Absorption Diseases of the stomach, small intestine, biliary tree, and pancreas can disrupt digestion and absorption. The most common maldigestion syndrome, lactase deficiency, produces gas and diarrhea after ingesting dairy products but has no adverse outcomes. Intestinal sucrase-isomaltase deficiency produces similar symptoms after consuming sucrose. Disorders that affect digestion and/or absorption more diffusely, including celiac disease, bacterial overgrowth, infectious enteritis, Crohn's ileitis, and radiation enteritis produce anemia, dehydration, electrolyte disorders, or malnutrition. Gastric acid hypersecretory conditions impair digestion by damaging the intestinal mucosa, impairing pancreatic enzyme activation, and accelerating transit. Benign or neoplastic biliary obstruction impairs fat digestion. Impaired pancreatic enzyme release in chronic pancreatitis or pancreatic cancer impairs digestion and leads to malnutrition.

Altered Secretion Some GI diseases result from alteration of gut secretion. Gastric acid hypersecretion occurs in gastrinoma, G-cell hyperplasia, retained antrum, and some patients with duodenal ulcers. Gastric acid is reduced in atrophic gastritis and pernicious anemia. Intestinal or colonic hypersecretion and/or impaired absorption causes diarrhea with acute bacterial or viral infection, chronic *Giardia* or cryptosporidia infections, small-intestinal bacterial overgrowth, bile salt diarrhea, microscopic colitis, diabetic diarrhea, and endocrine neoplasias with tumor overproduction of secretagogue transmitters such as vasoactive intestinal polypeptide.

Altered Gut Transit Mechanical obstruction can block transit of gut contents. Esophageal occlusion most often is due to stricture (from acid exposure or eosinophilic esophagitis) or neoplasm. Gastric obstruction develops from ulcer disease or cancer. Small-intestinal obstruction commonly results from adhesions but also occurs with Crohn's disease, radiation- or drug-induced strictures, and malignancy. The most common cause of colonic obstruction is colon cancer, although inflammatory strictures develop with inflammatory bowel disease (IBD), after diverticulitis, or with some medications.

Delayed propulsion can develop in any gut region. Achalasia is characterized by impaired esophageal body peristalsis and incomplete lower esophageal sphincter relaxation. Gastroparesis is the delay in gastric emptying resulting from impaired antral motility or spastic

pyloric contraction. Intestinal pseudoobstruction is the disruption of small-bowel contractility due to enteric nerve or smooth-muscle injury. Slow-transit constipation results from diffusely impaired colon propulsion. Constipation also occurs with anorectal outlet abnormalities such as rectal prolapse, intussusception, or dyssynergia—a failure of anal or puborectalis relaxation with defecation.

Disorders of rapid propulsion are less common. Rapid gastric emptying occurs with postvagotomy dumping syndrome and some cases of functional dyspepsia and cyclic vomiting syndrome. Exaggerated intestinal or colonic motor patterns may contribute to diarrhea in irritable bowel syndrome (IBS). Accelerated transit with hyperdefecation is noted in hyperthyroidism.

Immune Dysregulation Many inflammatory GI conditions are consequences of altered immune function. Mucosal inflammation in celiac disease results from dietary ingestion of gluten-containing grains. Eosinophilic esophagitis and eosinophilic gastroenteritis exhibit prominent mucosal eosinophil infiltration. Ulcerative colitis and Crohn's disease are types of IBD that produce mucosal injury primarily in the lower gut. The microscopic colitides, lymphocytic and collagenous colitis, exhibit subepithelial infiltrates without visible mucosal damage. Abnormal mast cell function underlies mastocytosis and may play a role in some IBS cases. Bacterial, viral, and protozoal organisms produce ileitis or colitis in selected patients. GI manifestations of COVID-19 include nausea, vomiting, diarrhea, abdominal pain, and hepatic abnormalities. Gut microbiome alterations (termed dysbiosis) may trigger IBD, celiac disease, and IBS flares and may be factors in oncogenesis in some cases of pancreatic cancer.

Impaired Gut Blood Flow Ischemic damage occurs as a consequence of impaired blood flow. Rare cases of gastroparesis result from celiac artery blockage. Intestinal and colonic ischemia are more common and can be consequences of arterial embolus or thrombosis, venous thrombosis, or hypoperfusion from dehydration, sepsis, hemorrhage, or reduced cardiac output. These conditions produce mucosal injury, hemorrhage, perforation, or stricturing with chronic ischemia. Radiation enterocolitis may lead to reduced mucosal blood flow without large vessel occlusion.

Neoplastic Degeneration All GI regions are susceptible to malignant degeneration. In the United States, colorectal cancer is most common and usually presents after age 45. Gastric cancer is prevalent, especially in certain Asian populations, and may be a consequence of *Helicobacter pylori* infection. Esophageal cancer develops with chronic acid reflux or after excessive alcohol or tobacco use. Small-intestinal neoplasms are rare but occur with underlying inflammatory diseases. Most GI cancers exhibit carcinomatous histology, but lymphomas and other cell types can be seen. Anal cancers arise after prior anal infection or inflammation. Pancreatic and biliary cancers elicit severe pain, weight loss, and jaundice and have poor prognoses. Hepatocellular carcinoma usually arises in the setting of chronic viral hepatitis or cirrhosis secondary to other causes.

Disorders without Obvious Organic Abnormalities The most prevalent GI disorders show no abnormalities on biochemical or structural testing and include IBS, functional dyspepsia, and functional heartburn. These disorders have been given a new nomenclature and are termed disorders of gut-brain interaction (DGBI). DGBIs exhibit complex pathophysiology including altered gut motor function, exaggerated visceral sensory response to noxious stimulation, abnormal central nervous system processing of visceral pain sensation, and mucosal immune defects. Many patients with DGBIs have significant emotional disturbances on psychometric testing.

Genetic Influences Some GI diseases exhibit hereditary components. Family members of IBD patients show genetic predisposition to disease development. Colonic, esophageal, and pancreatic malignancies arise in certain inherited disorders. Lynch syndrome is an autosomal dominant condition with increased risk for developing colon and gynecologic cancers. Genetic dysmotility syndromes are described. Familial clustering is observed in DGBIs, although this may be secondary learned familial illness behavior rather than a true genetic factor.

SYMPTOMS OF GI DISEASE

Symptoms of GI disease include abdominal pain, heartburn, nausea and vomiting, altered bowel habits, GI bleeding, jaundice, and other manifestations (Table 332-1).

Abdominal Pain Abdominal pain results from GI disease and extraintestinal conditions of the genitourinary tract, abdominal wall, thorax, or spine. Visceral pain generally is midline in location and vague in character. Parietal pain is localized and precisely described. Painful inflammatory diseases include peptic ulcer, appendicitis, diverticulitis, IBD, pancreatitis, cholecystitis, and infectious enterocolitis. Noninflammatory sources include biliary colic, mesenteric ischemia, and neoplasia. The most common causes of abdominal pain are IBS and functional dyspepsia.

Heartburn Heartburn, a burning substernal sensation, is experienced by 40% of the population. Heartburn usually results from excess gastroesophageal acid reflux, but some cases exhibit normal acid exposure and are caused by nonacidic reflux or heightened esophageal nerve sensitivity.

Nausea and Vomiting Nausea and vomiting are caused by GI diseases, medications, toxins, infection, endocrine disorders, labyrinthine conditions, and central nervous system disease. Mechanical obstruction should be excluded as a cause of chronic nausea and vomiting. Disorders of propulsion including gastroparesis and intestinal pseudoobstruction elicit similar symptoms. Nausea and vomiting also are reported with IBS or gastroduodenal DGBIs (chronic nausea vomiting syndrome, cyclic vomiting syndrome, cannabinoid hyperemesis syndrome).

Altered Bowel Habits Altered bowel habits are common complaints. Constipation may be reported as infrequent defecation,

TABLE 332-1 Common Causes of Common Gastrointestinal (GI) Symptoms

ABDOMINAL PAIN	NAUSEA AND VOMITING	DIARRHEA	GI BLEEDING	OBSTRUCTIVE JAUNDICE
Appendicitis	Medications	Infection	Ulcer disease	Bile duct stones
Gallstone disease	GI obstruction	Poorly absorbed sugars	Esophagitis	Cholangiocarcinoma
Pancreatitis	Motor disorders	Inflammatory bowel disease	Varices	Cholangitis
Diverticulitis	Functional bowel disorder	Microscopic colitis	Vascular lesions	Sclerosing cholangitis
Ulcer disease	Cyclic vomiting syndrome	Functional bowel disorder	Neoplasm	Ampullary stenosis
Esophagitis	Cannabinoid hyperemesis syndrome	Celiac disease	Diverticula	Ampullary carcinoma
GI obstruction	Enteric infection	Pancreatic insufficiency	Hemorrhoids	Pancreatitis
Inflammatory bowel disease	Pregnancy	Hyperthyroidism	Fissures	Pancreatic tumor
Functional bowel disorder	Endocrine disease	Ischemia	Inflammatory bowel disease	
Vascular disease	Motion sickness	Endocrine tumor	Infectious colitis	
Gynecologic causes	Central nervous system disease			
Renal stone				

straining with defecation, passage of hard stools, or a sense of incomplete fecal evacuation and is caused by obstruction, motor disorders, medications, and endocrine diseases (hypothyroidism, hyperparathyroidism). Diarrhea may be reported as frequent defecation, passage of loose/watery stools, fecal urgency, or a similar sense of incomplete evacuation and can result from infections, inflammatory causes, malabsorption, and medications. IBS produces constipation, diarrhea, or an alternating bowel pattern. Fecal mucus is common in IBS, whereas pus and blood characterize IBD. Malabsorption can cause steatorrhea. A history of contaminated water exposure raises concern for protozoal causes (*Giardia, Cryptosporidium*).

GI Bleeding Upper GI bleeding presents with melena or hematemesis, whereas lower GI bleeding produces bright red or maroon stools. However, briskly bleeding upper sites can elicit red rectal bleeding, while slowly bleeding ascending colon sites may cause melena. Chronic occult GI bleeding may present with iron deficiency anemia. Causes of upper GI bleeding include ulcers, gastroduodenitis, esophagitis, portal hypertensive etiologies, malignancy, gastroesophageal junction tears, and vascular lesions. Lower GI sources include hemorrhoids, anal fissures, diverticula, ischemic colitis, neoplasm, IBD, infectious colitis, drug-induced colitis, arteriovenous malformations, and other vascular lesions.

Jaundice Jaundice results from prehepatic, intrahepatic, or posthepatic disease. Posthepatic causes of jaundice include biliary (choledocholithiasis, acute cholangitis, primary sclerosing cholangitis, strictures, neoplasm) and pancreatic (pancreatitis, stricture, malignancy) disorders.

Other Symptoms Dysphagia, odynophagia, and unexplained chest pain suggest esophageal disease. Globus sensation is reported with acid reflux but also occurs with upper esophageal sphincter dysfunction, hypersensitivity, or affective disorders. Weight loss, anorexia, and fatigue are reported with neoplastic, inflammatory, motility, pancreatic, and psychiatric conditions. IBD is associated with hepatobiliary dysfunction, skin and eye lesions, and arthritis. Celiac disease may present with dermatitis herpetiformis. Jaundice produces pruritus. Conversely, systemic diseases have GI consequences. Systemic lupus may cause gut ischemia, presenting with pain or bleeding.

EVALUATION OF THE PATIENT WITH GI DISEASE

Evaluation of suspected GI disease begins with a careful history and examination and may be followed by investigation with tools to test gut structure or function and luminal contents in selected cases. In patients with normal diagnostic assessment, validated symptom profiles can be characterized to confidently define the presence of a DGBI.

HISTORY

The history in suspected GI disease has several components. Symptom timing, profiles, and duration suggest specific etiologies. Short-duration symptoms commonly result from acute infection or inflammation, toxin exposure, or ischemia. Long-standing symptoms point to chronic inflammation, neoplasia, or DGBIs. Luminal obstruction presents with dysphagia, nausea and vomiting, bloating and distention, or constipation depending on the site of blockage. Symptoms from obstruction, ischemia, IBD, and DGBIs are worsened by meals, while ulcer symptoms may be relieved by eating. Ulcer pain occurs intermittently over weeks to months, whereas biliary colic has sudden onset and lasts up to several hours. Acute pancreatitis pain is severe and persists for days to weeks. Meals elicit diarrhea while defecation relieves discomfort in some IBD and IBS cases. DGBIs are exacerbated by stress. Sudden awakening by pain suggests organic rather than functional causes. Diarrhea from malabsorption usually improves with fasting, whereas secretory diarrhea persists without oral intake.

Symptom relation to other factors offers insight into underlying diagnostic possibilities. Obstructive symptoms with a history of abdominal surgery raise concern for adhesions. Loose stools after gastrectomy or cholecystectomy suggest dumping syndrome or postcholecystectomy diarrhea. Symptom onset after travel prompts consideration of infection. Medications produce pain, altered bowel habits, or GI bleeding. Celiac disease is prevalent in people of northern European descent, whereas IBD is more common in Jewish populations. A sexual history may raise concern for infection or immunodeficiency.

Symptom criteria devised by working groups improve diagnosis of DGIBs and reduce unnecessary diagnostic testing. The best accepted symptom-based criteria are the Rome criteria, which exhibit sensitivities and specificities of 55–75% when tested against structural findings in IBS and functional dyspepsia, indicating the need for careful test selection in patients with high risk of organic disease.

PHYSICAL EXAMINATION

The physical examination complements information from the history. Fever suggests inflammation or neoplasm. Orthostasis is produced by significant blood loss, dehydration, sepsis, or autonomic neuropathy. Skin, eye, or joint findings may point to specific diagnoses such as IBD. Neck examination with swallowing assessment evaluates dysphagia. Lung and cardiac examinations can suggest cardiopulmonary disease as causes of abdominal pain or nausea, while pelvic examination tests for gynecologic sources of pain. Rectal examination may detect blood from mucosal injury or neoplasm or a palpable inflammatory mass in appendicitis. Metabolic conditions and gut motor disorders can have associated peripheral neuropathy.

Abdominal inspection may reveal distention from obstruction, neoplasm, or ascites or vascular abnormalities with liver disease. Auscultation detects bruits or friction rubs from vascular disease or hepatic tumors. Loss of bowel sounds signifies ileus, whereas high-pitched, hyperactive sounds characterize intestinal obstruction. Percussion assesses liver size and detects shifting dullness from ascites. Palpation assesses for hepatosplenomegaly and neoplastic or inflammatory masses. Intestinal ischemia elicits severe pain but little tenderness. Patients with visceral pain may exhibit generalized discomfort, whereas those with parietal pain or peritonitis have localized pain with involuntary guarding, rigidity, or rebound. Patients with musculoskeletal abdominal wall pain may note tenderness exacerbated by Valsalva or leg lift maneuvers.

TOOLS FOR PATIENT EVALUATION

Laboratory, radiographic, and functional tests help to diagnose suspected GI disease. The GI tract is amenable to internal evaluation using endoscopy and to examination of luminal contents. Histopathologic examination of GI tissues complements this testing.

Laboratory Laboratory tests can provide useful information. Iron-deficiency anemia suggests mucosal blood loss, whereas vitamin B_{12} deficiency results from intestinal, gastric, or pancreatic disease. Both conditions also result from inadequate oral intake. Leukocytosis and increased sedimentation rates and C-reactive proteins are found in inflammation, whereas leukopenia is seen in viral illness. Severe vomiting or diarrhea elicits electrolyte disturbances, acid-base abnormalities, and elevated blood urea nitrogen. Pancreaticobiliary or liver disease produces elevated pancreatic or liver chemistries. Thyroid chemistries, cortisol, and calcium levels evaluate for endocrinologic causes of symptoms. Pregnancy testing is considered for women with unexplained nausea. Serologic tests screen for atrophic gastritis, celiac disease, connective tissue diseases, and paraneoplastic dysmotility syndromes. Hormone and neurotransmitter levels are obtained for suspected endocrine neoplasia. Measurement of serum gastrin after secretin administration can facilitate gastrinoma diagnosis. Intraabdominal malignancies produce tumor markers including carcinoembryonic antigen, CA 19-9, CA 125, and α-fetoprotein. Blood testing monitors thiopurine metabolites in IBD patients on immunosuppressant agents. Research into novel biomarkers is being conducted to predict longitudinal course and treatment response in IBD and other conditions. Ascitic fluid is analyzed for infection, malignancy, or findings of portal hypertension. Urine samples screen for carcinoid, porphyria, and heavy metal intoxication. Genetic testing is indicated in some patients with hereditary predisposition to selected GI diseases. Pharmacogenetic methods can determine which therapies should not be given to specific patients.

Luminal Contents Luminal contents are analyzed in selected settings. Fecal *Helicobacter pylori* antigen testing can detect active infection with this agent in ulcer disease or dyspepsia. Stool samples are cultured for bacterial pathogens, examined for leukocytes and parasites, or tested for *Giardia* antigen. Fecal fat is increased with malabsorption. Fecal elastase is decreased with exocrine pancreatic insufficiency. Inflammatory conditions such as IBD produce fecal calprotectin or lactoferrin elevations. Stool electrolytes can be measured in diarrheal conditions. Laxative screens are performed for suspected laxative abuse. Fecal immunochemical and multitarget fecal DNA tests can be used for colon cancer screening in low-risk populations. Gastric acid is quantified to exclude gastrinoma. Esophageal pH/impedance testing is done for refractory gastroesophageal reflux and can characterize mechanisms underlying unexplained belching.

Endoscopy Indications for endoscopic procedures are listed in Table 332-2, which can diagnose causes of bleeding, pain, nausea and vomiting, weight loss, altered bowel function, and fever. Upper endoscopy evaluates the esophagus, stomach, and duodenum and is advocated as the initial test for suspected ulcer disease, esophagitis, neoplasm, malabsorption, and Barrett's metaplasia and dysplasia because of its abilities to visualize and biopsy abnormalities. Endoscopically obtained duodenal aspirates can be examined for parasites or cultured for bacterial overgrowth. It should be recognized that upper endoscopy is normal in 85% of patients with unexplained dyspepsia. Colonoscopy is the preferred procedure for colon cancer screening and surveillance with possible excision of noncancerous polyps. Colonoscopy also can obtain biopsies for colitis or ileitis secondary to IBD, microscopic colitis, immune checkpoint inhibitor use, infection, ischemia, and radiation. Sigmoidoscopy examines the colon to the splenic flexure and excludes distal causes of bleeding, inflammation, or obstruction in young patients not at significant risk for colon cancer. For elusive GI bleeding from arteriovenous malformations or superficial ulcers, small-intestinal examination is performed with push enteroscopy, capsule endoscopy, or double-balloon enteroscopy. Capsule endoscopy also visualizes small-intestinal Crohn's disease in individuals with negative radiography. Endoscopic ultrasound (EUS) diagnoses and stages GI malignancy, excludes choledocholithiasis, evaluates pancreatitis, and assesses anal sphincter integrity. Endoscopic retrograde cholangiopancreatography (ERCP) provides diagnoses and affords therapy of pancreatic and biliary disease.

Radiography/Nuclear Medicine Radiographic tests evaluate visceral and extraluminal structures. Esophagography is the initial procedure to exclude subtle rings, strictures, or achalasia as causes of dysphagia, whereas fluoroscopic triple-phase swallow testing evaluates oropharyngeal dysphagia. Small-bowel contrast radiology detects intestinal tumors, strictures, and fistulae and can estimate intestinal transit. Contrast enemas are performed when colonoscopy is unsuccessful or contraindicated. Defecography characterizes anorectal dysfunction as a cause of constipation. Ultrasound and computed tomography (CT) evaluate regions not accessible by endoscopy or contrast studies, including the liver, pancreas, gallbladder, kidneys, and retroperitoneum, and can diagnose appendicitis, mass lesions, fluid collections, organ enlargement, and (in the case of ultrasound) gallstones. CT and magnetic resonance (MR) colonography are alternatives to colonoscopy for colon cancer screening. MR methods image pancreaticobiliary ducts to exclude neoplasm, stones, and sclerosing cholangitis and the liver to characterize benign and malignant tumors. CT or MR enterography assesses small-bowel thickening, stricturing, masses, and fistulae in IBD and other inflammatory or neoplastic conditions. Percutaneous transhepatic cholangiography accesses the biliary tree in obstructive jaundice. CT and MR angiographic techniques can define mesenteric occlusion and detect acute intraluminal hemorrhage. Ultrasound or MR elastography quantifies fibrosis in chronic liver disease. Positron emission tomography (PET) distinguishes malignant from benign disease in several organ systems. Imaging with combined PET-CT techniques using DOTA-octreotate and related agents has improved detection of neuroendocrine tumors.

Scintigraphy evaluates structural abnormalities and quantifies luminal transit. Radionuclide scans localize bleeding sites in patients with brisk hemorrhage to direct therapy with endoscopy, angiography, or surgery. Radiolabeled leukocyte scans search for intraabdominal abscesses not visualized on CT. Biliary scintigraphy complements ultrasound in assessing cholecystitis. Scintigraphy to quantify esophageal and gastric emptying is well established. Scintigraphic measures of small-intestinal or colonic transit are less widely used.

TABLE 332-2 Common Indications for Endoscopy

UPPER ENDOSCOPY	COLONOSCOPY	ENDOSCOPIC RETROGRADE CHOLANGIOPANCREATOGRAPHY	ENDOSCOPIC ULTRASOUND	CAPSULE ENDOSCOPY	DOUBLE-BALLOON ENDOSCOPY
Dyspepsia despite treatment	Cancer screening	Jaundice	Staging of malignancy	Obscure bleeding	Ablation of small-intestinal bleeding sources
Dyspepsia with signs of organic disease	Lower gastrointestinal (GI) bleeding	Postbiliary surgery complaints	Characterize and biopsy submucosal mass	Suspected Crohn's disease of the small intestine	Biopsy of suspicious small-intestinal masses/ulcers
Refractory vomiting	Anemia	Cholangitis	Bile duct stones		
Dysphagia	Diarrhea	Gallstone pancreatitis	Chronic pancreatitis		
Upper GI bleeding	Polypectomy	Pancreatic/biliary/ampullary tumor	Drain pseudocyst		
Anemia	Obstruction	Unexplained pancreatitis	Anal continuity		
Weight loss	Biopsy radiologic abnormality	Pancreatitis with unrelenting pain	Direct stent placement		
Malabsorption	Cancer surveillance: family history prior polyp/cancer, colitis	Fistulas			
Biopsy radiologic abnormality	Palliate neoplasm	Biopsy radiologic abnormality			
Polypectomy	Remove foreign body	Pancreaticobiliary drainage			
Place gastrostomy	Place stent across stenosis	Sample bile			
Barrett's surveillance		Sphincter of Oddi manometry			
Palliate neoplasm					
Sample duodenal tissue/fluid					
Remove foreign body					
Endoscopic mucosal resection or endoscopic submucosal dissection for dysplasia or early cancer					
Place stent across stenosis					
Endoscopic myotomy for achalasia or gastroparesis					
Endoscopic bariatric procedures					

Histopathologic Assessment Endoscopic mucosal biopsies evaluate for inflammatory (including celiac disease, IBD, microscopic colitis), infectious, dysplastic, and neoplastic disease. Gastric biopsies can detect *H. pylori* on immunohistochemical staining and determine antibiotic sensitivity from biopsy cultures. Deep rectal biopsies facilitate diagnosis of Hirschsprung's disease or amyloid. Liver biopsy is performed for abnormal liver chemistries, unexplained jaundice, some cases of viral hepatitis, and following liver transplants to exclude rejection. Biopsies obtained during CT or ultrasound evaluate for intraabdominal conditions not accessible by endoscopy.

Methods to visually scrutinize histologic features and detect dysplasia during endoscopy prior to targeted biopsy in colitis, Barrett's esophagus, and gastric malignancy include narrow-band imaging, chromoendoscopy, confocal laser endomicroscopy, and optical coherence tomography. Artificial intelligence techniques using machine learning are in development for detecting dysplasia and early cancer in images from biopsies.

Functional Testing Tests of gut function provide important data when structural testing is nondiagnostic. High-resolution esophageal manometry defines esophageal motility disorders including achalasia. Small-intestinal manometry tests for pseudoobstruction and colon manometry evaluate for colonic inertia. Anorectal manometry with balloon expulsion testing characterizes unexplained incontinence or constipation from outlet dysfunction. The functional lumen imaging probe measures reduced distensibility in the lower esophageal sphincter in achalasia, the upper esophageal sphincter with Zenker's diverticula, and the pylorus in gastroparesis. Measurement of breath hydrogen and methane while fasting and after oral mono- or oligosaccharide challenge screen for carbohydrate intolerance and small-intestinal bacterial overgrowth. Urea breath testing assesses for persistent *H. pylori* infection. Gastric emptying breath testing is an alternative to scintigraphy for gastroparesis diagnosis.

TREATMENT

Gastrointestinal Disease

Management of GI diseases depends on the cause of symptoms. Treatment options include dietary modifications, medications, treatment of gut dysbiosis, therapeutic endoscopy, interventional radiology, surgery, psychological approaches, and physical therapy. Smartphone applications have been developed for diverse purposes ranging from providing instructions for endoscopy preparation to educating and promoting adherence to diet restrictions.

NUTRITIONAL MODIFICATION

Nutritional modifications for GI disease can reduce symptoms, correct pathologic defects, or replace normal food intake with enteral or parenteral formulations. Changes that improve symptoms but do not reverse organic abnormalities include lactose restriction for lactase deficiency, liquid meals in gastroparesis, carbohydrate restrictions with dumping syndrome, avoidance of selected meats with alpha gal syndrome, and low-FODMAP (fermentable oligo-, di-, monosaccharides and polyols) diets in IBS. The gluten-free diet for celiac disease exemplifies a primary therapy to reduce mucosal inflammation. Likewise, elimination diets may improve histology and symptoms in eosinophilic esophagitis. Medium-chain triglycerides replace normal fats in short-gut syndrome or severe ileal disease. Perfusing liquid meals through a gastrostomy is performed in patients who cannot swallow safely. Enteral jejunostomy feedings are considered for gastric dysmotility syndromes that preclude feeding into the stomach. Total parenteral hyperalimentation is used for generalized gut failure in patients for whom enteral nutrition cannot be sustained.

PHARMACOTHERAPY

Medications in several categories treat GI diseases. Considerable resources are expended on over-the-counter remedies. Prescription drugs are offered as short-term or continuous therapy to reduce GI symptoms. Other prescription agents modify disease activity with the intent to alter longitudinal disease course. Alternative treatments are popular in conditions for which traditional therapies provide incomplete relief.

Over-the-Counter Agents Over-the-counter agents are reserved for mild GI symptoms. Antacids, histamine H_2 antagonists, alginates, and proton pump inhibitors (PPIs) decrease symptoms in gastroesophageal reflux disease (GERD) and dyspepsia. Fiber supplements, stool softeners, enemas, and laxatives are used for constipation. Laxatives are categorized as stimulants, osmotic agents (including isotonic preparations containing polyethylene glycol), and poorly absorbed sugars. Nonprescription antidiarrheal agents include bismuth subsalicylate, kaolin-pectin combinations, and loperamide. Lactase pills are used for lactose intolerance. Gaseous symptoms may be reduced by bacterial α-galactosidase, antiflatulents, and adsorbents. Using nonprescription preparations for chronic persistent symptoms should be supervised by a health care provider to ensure that an underlying severe medical illness is not missed.

Prescription Drugs for Symptom Control Prescription drugs are approved to treat symptoms in many GI diseases. Higher-dose PPIs are advocated for GERD when over-the-counter preparations are inadequate. Given acutely, PPIs slow acute upper GI bleeding, reducing requirements for endoscopic therapy. PPIs slow progression of Barrett's metaplasia and can manage eosinophilic esophagitis. Vonoprazan is the first drug in a newer acid suppressant class of potassium-competitive acid blockers for severe acid-related disorders and as an adjunct to *H. pylori* eradication. Baclofen reduces gastroesophageal reflux by increasing lower esophageal sphincter barrier function. Other therapies of eosinophilic esophagitis include swallowed fluticasone and budesonide. Cytoprotective agents like sucralfate are available for upper gut ulcers. Prokinetic drugs stimulate GI propulsion in gastroparesis, chronic intestinal pseudoobstruction, acute colonic pseudoobstruction, and slow-transit constipation. Secretagogue drugs are prescribed for refractory constipation, whereas peripheral opioid antagonists are offered for opioid-induced constipation. Prescription antidiarrheals include opioid drugs, anticholinergic antispasmodics, tricyclics, bile acid binders, and serotonin antagonists. Opioids may be needed for pain control in disseminated malignancy and chronic pancreatitis. Antiemetics reduce nausea and vomiting. Pancreatic enzymes decrease malabsorption and pain from pancreatic disease. The antisecretory somatostatin analogue octreotide treats hypersecretory states. Some DGBIs require neuromodulators, including tricyclic agents, for control of pain, diarrhea, or nausea. Nonselective beta blockers can reduce the risk of variceal hemorrhage with portal hypertension.

Prescription Drugs for Disease Modification Other prescription medication classes modify disease activity by actions to reduce inflammation, modulate immune function, and target neoplasm. Immunomodulatory drugs are used in IBD, microscopic colitis, immune checkpoint inhibitor colitis, refractory celiac disease, autoimmune pancreatitis, and gut vasculitis. Biologic agents that inhibit tumor necrosis activity and antibodies targeting integrins, proinflammatory cytokines, Janus kinase inhibitors, and cell adhesion molecules have benefits in IBD. Agents effective for acute severe ulcerative colitis requiring hospitalization include corticosteroids, infliximab, and cyclosporine. Biologics that deplete eosinophils or inhibit mast cells show promise in eosinophilic disorders. *H. pylori* eradication treats some gastric malignancies. Most GI carcinomas respond poorly to chemotherapy with or without radiotherapy, whereas lymphomas may be cured with these interventions. Cancer treatment programs including immunotherapies, immune checkpoint inhibitors, and tumor vaccines are in development.

Complementary and Alternative Medicines Several alternative treatments are available to manage GI symptoms. Ginger, acupressure, and acustimulation are used for nausea. Pyridoxine is given